

Devi Ahilya University, Institute of Engineering & Technology
Class Test: 1, AUG' 23
3EIRC2: Digital Electronics
II Year Electronics & Instrumentation Engg.

Maximum Marks: 20

Time: 70 Minutes

Q1. Show all conversion steps for below

i. Express (53.75)₁₀ in binary form.

ii. Express above obtained binary result in Octal and Hexadecimal form.

Simplify using Boolean laws

$$\overline{A}\overline{B} + ABC + A(B + A\overline{B})$$

Q2. Draw RTL logic circuit for 3-input NAND gate and explain its working.

Q3. Design gated circuit for the function

$$Z = \overline{B}D + \overline{B}C + \overline{A}\overline{B}C\overline{D} \text{ using Only NAND gates}$$

Simplify above expression and implement using AND-OR-INVERT logic gates.

Q4. It is needed to compare the magnitude of two 2-bit numbers A & B. Three conditions to be tested A = B, A < B using three different coloured LED's. Give logic expressions for every condition to be tested and, draw the gated circuit.